A framework for bilevel optimization that enables stochastic and global variance reduction algorithms

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1. Problem Statement and applications

Bilevel Optimization Problems

Let $F, G : \mathbb{R}^p \times \mathbb{R}^d \to \mathbb{R}$

- Goal:

$$\min_{x\in\mathbb{R}^d} h(x) \triangleq F(z^*(x), x), \text{ s.t. } z^*(x) \in \arg\min_{z\in\mathbb{R}^p} G(z, x) \ .$$

- ERM setting:

$$F(z,x) = \frac{1}{m}\sum_{j=1}^{m} F_j(z,x), \quad G(z,x) = \frac{1}{n}\sum_{i=1}^{n} G_i(z,x)$$

Examples of applications

- Hyperparameter selection [10]
- Deep Equilibrium Models [2]
- Neural architecture search [9]
- Data Augmentation [11]

2. First order methods

Gradient descent

$$x^{t+1} = x^t - \gamma \nabla h(x^t)$$

Implicit differentiation

$$\nabla h(x) = \nabla_2 F(z^*(x), x) + \nabla^2_{21} G(z^*(x), x) v^*(x)$$

with

$$v^*(x) = -[\nabla^2_{11} G(z^*(x), x)]^{-1} \nabla_1 F(z^*(x), x)$$

Bottlenecks

- One optimization problem to solve at each iteration
- One linear system to solve at each iteration
- Computation of the full batch derivatives when m and n are large

3. General framework

Idea: Maintain the variables z, v, x by alternating steps in the following directions

- $D_z(z,v,x) = \nabla_1 G(z,x)$: gradient step towards $z^*(x)$
- $D_v(z,v,x) = \nabla^2_{11} G(z,x)v + \nabla_1 F(z,x)$: gradient step towards $v^*(x)$
- $D_x(z,v,x) = \nabla^2_{21} G(z,x)v + \nabla_2 F(z,x)$: gradient step towards x^*

Input: initializations $z_0 \in \mathbb{R}^p$, $x_0 \in \mathbb{R}^d$, $v_0 \in \mathbb{R}^p$, number of iterations T, step size sequences $(\rho^t)_{t<T}$ and $(\gamma^t)_{t<T}$.

for $t = 0, \dots, T-1$ **do**

Update z: $z^{t+1} = z^t - \rho^t D^t_z$,
Update v: $v^{t+1} = v^t - \rho^t D^t_v$,
Update x: $x^{t+1} = x^t - \gamma^t D^t_x$,
where D^t_z, D^t_v and D^t_x are unbiased estimators of $D_z(z^t, v^t, x^t), D_v(z^t, v^t, v^t)$ and $D_x(z^t, v^t, x^t)$.

end

Output: (z^T, v^T, x^T)

4. StOchastic Bilevel Algorithm (SOBA)

Chosen directions

Pick $i \in [n]$ and $j \in [m]$ and take

- $D^t_z = \nabla_1 G_i(z^t, x^t)$
- $D^t_v = \nabla^2_{11} G_i(z^t, x^t) v^t + \nabla_1 F_j(z^t, x^t)$
- $D^t_x = \nabla^2_{21} G_i(z^t, x^t) v^t + \nabla_2 F_j(z^t, x^t)$

Convergence rate

$$\inf_{t\le T} \mathbb{E}[\|\nabla h(x^t)\|^2] = \mathcal{O}(T^{-\frac{2}{5}})$$

5. Aside: SAGA for single level problem [3]

Single level problem

$$\min_{x\in\mathbb{R}^p} f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$$

Initialization Compute and store $m[i] = \nabla f_i(x^0)$ for any $i \in \{1, \dots, n\}$ and $S[m] = \frac{1}{n}\sum_{i=1}^{n} m[i]$.

At iteration t:

1. Pick $i \in \{1, \dots, n\}$
2. Update x

$$x^{t+1} = x^t - \rho(\nabla f_i(x^t) \underbrace{- m[i] + S[m]}_{\text{variance reduction}})$$

3. Update the memory

$$m[i] \leftarrow \nabla f_i(x^t)$$

6. Stochastic Averaged Bilevel Algorithm (SABA)

Chosen directions

Estimate the five quantities $\nabla_1 G(z^t, x^t)$, $\nabla_1 F(z^t, x^t)$, $\nabla_2 F(z^t, x^t)$, $\nabla^2_{12} G(z^t, x^t) v^t$, $\nabla^2_{11} G(z^t, x^t) v^t$ on the principle of SAGA and plug these estimates in $D_z(z^t, v^t, x^t)$, $D_v(z^t, v^t, x^t)$, $D_x(z^t, v^t, x^t)$.

Convergence rate

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\|\nabla h(x^t)\|^2] = \mathcal{O}(T^{-1})$$

6. Application to hyperparameter selection

Setting

- Task: binary classification
- IJCNN1 dataset: $49\,990$ training samples, $91\,701$ validation samples, 22 features
- Training loss:

$$G(\theta, \lambda) = \frac{1}{n}\sum_{i=1}^{n} \log(1 + \exp(-y_i \langle x_i, \theta\rangle) + \frac{1}{2}\sum_{k=1}^{p} e^{\lambda_k}\theta_k^2$$

- Validation loss: logistic loss

$$F(\theta, \lambda) = \frac{1}{m}\sum_{j=1}^{m} \log(1 + \exp(-y_i^{\text{val}} \langle x_i^{\text{val}}, \theta\rangle)$$

- Competitor methods: BSA [4], TTSA [5], stocBiO [7], amIGO [1], MRBO [12], FSLA [8], SUSTAIN [6]

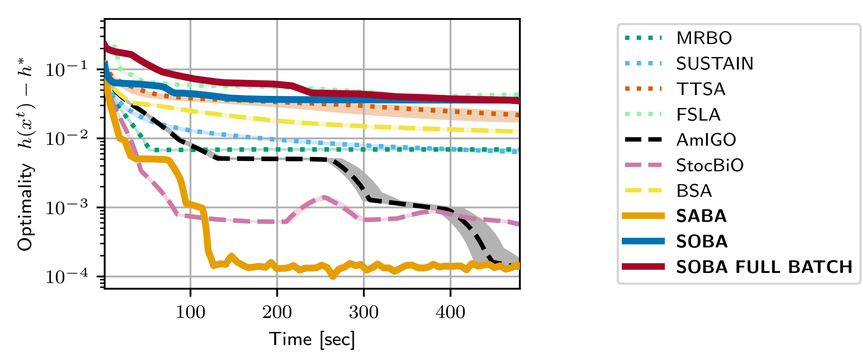

Figure: Comparison of SOBA and SABA with other stochastic bilevel optimization methods.

References

[1] M. Arbel, J. Mairal. Amortized Implicit Differentiation for Stochastic Bilevel Optimization. In *International Conference on Learning Representations (ICLR)*, 2022.
[2] S. Bai, J. Z. Kolter, V. Koltun. Deep Equilibrium Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
[3] A. Defazio, F. Bach, S. Lacoste-Julien. SAGA: A Fast Incremental Gradient Method With Support for Non-Strongly Convex Composite Objectives. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
[4] S. Ghadimi, M. Wang. Approximation Methods for Bilevel Programming. *arXiv 1802.02246*, 2018.
[5] M. Hong, M.-T. Wai, Z. Wang, Z. Yang. A Two-Timescale Framework for Bilevel Optimization: Complexity Analysis and Application to Actor-Critic. *arXiv 2007.05170*, 2021.
[6] P. Khanduri, S. Zeng, M. Hong, H.-T. Wai, Z. Wang, Z. Yang. A Near-Optimal Algorithm for Stochastic Bilevel Optimization via Double-Momentum. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
[7] K. Ji, J. Yang, Y. Liang. Bilevel optimization: Convergence analysis and enhanced design. In *International Conference on Machine Learning (ICML)*, 2021.
[8] J. Li, B. Gu, H. Huang, A Fully Single Loop Algorithm for Bilevel Optimization without Hessian Inverse. In *Proceedings of the Thirty-sixth AAAI Conference on Artificial Intelligence, AAAI'22*, 2022.
[9] H. Liu, K. Simonyan, Y. Yang. Darts: Differentiable ar-chitecture search. In *International Conference on Learn-ing Representations (ICLR)*, 2018.
[10] F. Pedregosa. Hyperparameter optimization with approximate gradient. In *International Conference on MachineLearning (ICML)*, 2016.
[11] C. Rommel, T. Moreau, J. Paillard, A. Gramfort. CADDA: Class-wise Automatic Differentiable Data Augmentation for EEG Signals, In *International Conference on Learning Representations (ICLR)*, 2022.
[12] J. Yang, K. Ji, Y. Liang. Provably Faster Algorithms for Bilevel Optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.